



BIBINA BOGDAN-IONUT

Software engineer

Work Experience

Alchemy Software engineer

2025 – 2026

Aptos Staking Indexer

Designed and implemented an **Aptos staking indexer** in Rust, leveraging the **Aptos RPC** to extract, aggregate, and analyze validator staking data — including **rewards, staking periods, delegator counts, and total staked amounts**. The system generated comprehensive validator performance statistics and insights stored in **PostgreSQL**, optimized for fast queries and scalability.

Built with libraries such as **reqwest** (for RPC calls), **serde** (for JSON serialization), **tokio** (for async concurrency), and **sqlx** (for database integration), the indexer featured a modular, production-ready architecture.

Freelancing Software engineer

2025 – 2026

Market Maker for Solana on multiple Dex and Cex

I developed an algorithmic trading system in Python, integrating the **Jupiter API, Binance API, and Solana RPC** to execute **automated trading strategies** across multiple decentralized and centralized exchanges (**DEXs** and **CEXs**). The system leveraged **multithreading** and **asynchronous processing** to achieve high-throughput order execution and low latency.

I built and managed several **trading bots** — including a **market-making engine for Solana liquidity pools** — each designed for real-time decision-making, and price optimization. The architecture emphasized scalability, fault tolerance, and performance tuning to ensure efficient execution in volatile crypto markets.

BwareLabs

2023 - 2025

Multi-Chains indexer

As part of a blockchain engineering team, I helped **design** and implement a scalable multi-chain **indexer capable** of tracking and serving data from **multiple networks** (e.g., **Ethereum, Arbitrum, Aptos**).

I built and maintained the **Python-based serve layer** from scratch, responsible for exposing indexed **blockchain** data via **REST** and **gRPC APIs**.

The system was structured into two main components:

- Indexer: handled on-chain data ingestion, event decoding, and block processing using **asynchronous** Python(Asyncio) and ethers-rs for **JSON-RPC** connections.
- Serve: made in python, in **FastAPI, managed querying, caching**, and API delivery, ensuring **high performance, fault tolerance**, and horizontal scalability.

Contributed to **system architecture** and **database schema design (PostgreSQL)**. Ensured production reliability through testing, monitoring, and optimization for large-scale blockchain data.

Contacts



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About Me

Results-driven **backend engineer** with 6+ years of experience developing high-performance web applications and **distributed systems**. Over the past **4+ years**, I've specialized in Rust, building scalable microservice **architectures** and **blockchain solutions** across **multiple chains**, including **Ethereum, Starknet, Solana, Aptos** and others. Skilled in **PostgreSQL**, performance optimization, and API design, I focus on delivering robust, maintainable, and production-ready solutions. Passionate about clean architecture and continuous improvement, I thrive in **cross-functional teams** that value innovation and collaboration.

Education

Bachelor's degree in Systems Engineering

Faculty of Automatic Control and Computer Science, UPB

2019 - 2023

Skills

Languages & Frameworks:

Rust, Python, SQL, HTML, CSS, JavaScript

Blockchain Technologies:

Ethereum, Arbitrum, Aptos, Solana, JSON-RPC, Web3, Ethers-rs, WebSocket APIs, Event Indexing, On-Chain Data, Governance Protocols, IPFS

Backend & Systems:

Asynchronous Programming, REST API, Microservices Architecture, Distributed Systems, High Availability, GraphQL, Object Oriented Programming, Git, Pytest, FastAPI, Celery

Databases & Storage:

PostgreSQL, SQL Optimization, Data Modeling, MongoDB, NoSQL, Redis

DevOps & Scalability:

Docker, Docker Compose, GitHub Actions, Monitoring, Logging, System Performance Optimization, Kubernetes, AWS, S3, R2, Cloudflare, DynamoDB, CI/CD pipelines, Lambda

Govscan - proposal voting platform

Built in **Rust**, GovScan is a **cross-chain proposal tracking** platform that indexes governance activity across multiple blockchains.

I implemented support for **Arbitrum** and **Ethereum**, creating modules to **index on-chain** events directly from the blockchain to extract proposal creation, voting, and execution data from smart contracts. Used ethers-rs and **async Rust** (Tokio) to stream and decode **event logs** via JSON-RPC/**WebSocket** connections, ensuring real-time updates of governance activity.

Integrated the **indexed data** into a **PostgreSQL** backend and exposed it through optimized FastAPI endpoints for analytics dashboards and external clients.

Python Monitoring Exporter

Developed a **Python-based monitoring** and **metrics exporter** for Bware Labs' main RPC infrastructure, enabling **real-time performance** visibility and reliability tracking. The exporter collected, aggregated, and exposed RPC metrics (latency, uptime, request rate, error rate, etc.) in a standardized format (e.g., **Prometheus-compatible endpoints**). It integrated with **Grafana** dashboards and internal **observability tools** to provide actionable insights and automated alerts for infrastructure reliability, **load balancing**, and **performance optimization**.

Python NFT Image Normalization Service

Developed a **Python service** for **automated image normalization** across **large batches** of **NFT assets** with inconsistent **dimensions, formats, and quality**. Implemented logic to read and analyze **image byte content**, detect image type and metadata, and apply **custom filters, resizing, and format standardization** for optimized on-chain and marketplace display. The system ensured uniform visual quality and metadata consistency across all **NFT collections**.

Bachelor's Thesis

Python Cross-Marketplace Price Comparison & Product Recommendation

- Built a **Chrome extension** that finds the closest **matching product** across multiple marketplaces and surfaces live price differences.
- Implemented **NLP keyword matching** with **Levenshtein distance** for **fuzzy string similarity** on titles/descriptions, plus text normalization.
- Designed a **KNN-based recommender** for similar items and added a **site-level penalization** heuristic to **down-rank** marketplaces with unreliable structure or noisy results.
- Created a fast marketplace **onboarding flow** (schema mapping, extraction rules, normalization) to add new sites quickly.
- Performance: **local caching, deduping requests**, and a composite ranking (**relevance + price delta + penalization**).